Some Blu-ray And HD-DVD specs

Parameter	Blu-Ray	HD-DVD
Capacity (ROM SL)	23.3/25 GB	15 GB
Laser wavelength	405 nm	405 nm
Numerical Aperture	0.85	0.65
Read power	0.35 mW	0.50 mW
Protection layer	0.1 mm	0.6 mm
Track pitch	0.32 µm	0.40 µm
Minimum pit length	~150 nm	204 nm

bumps and pits. A bump is an area of the substrate (the material on which the data is written) where there is no pit, so the pit is what is taken into consideration when one talks about burning data onto optical media. It is definitely not as simple as “a pit is a one and a bump is a zero,” but one can think of it that way to simplify things.

HD-DVD, too, uses the shorter-wavelength blue-violet laser. The name “HD-DVD” itself stands for, of course, High-Density Digital Versatile Disc.

Both the Blu-ray and HD-DVD camps say they will initially be providing “hybrid discs,” with a high-definition disc on one side and a regular DVD on the other—so that consumers will be able to play movies on their DVD players for now, in a measure of future-proofing.

How Are BD And HD-DVD Different?

We need to explain three terms here—numerical aperture (NA), track pitch, and pit length.

The light-gathering capacity of a lens is indicated by its NA. It is dependent on the diameter of the lens as well as on the quality of the optics. The higher the NA—meaning the lens can gather more light—the better.

The track pitch (see figure *Track Pitch And Pit Length*) is the distance between the centres of two successive rounds of the spiral track on which the pits are burnt. The smaller the track pitch, the more the number of tracks on the disc, and hence the more the data that can be stored on the disc as a whole.

The pit length (again, see figure *Track Pitch And Pit Length*) is simply the size of the burnt pit.

Now, the NA and the wavelength define the size of the laser beam. A higher NA and a shorter wavelength means a laser beam with a smaller area of illumination, which allows for the focusing of the beam with higher precision—and therefore, the possibility of a reduction in the track pitch and pit length.

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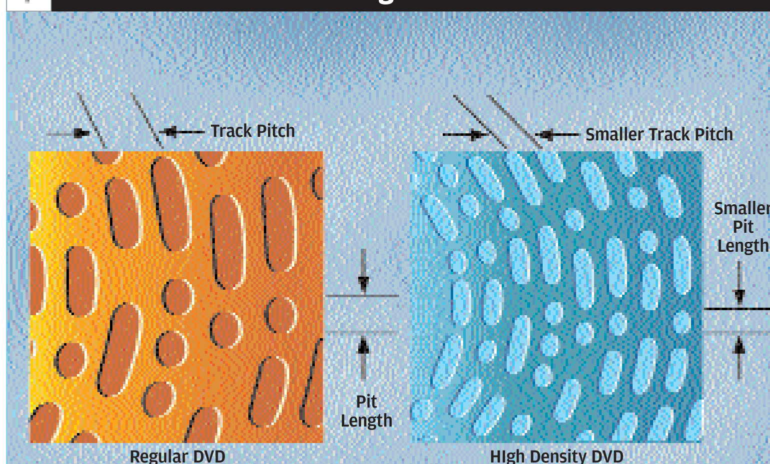
Where the technical specs are concerned, the main difference between Blu-ray and HD-DVD is in the disc structure itself. HD-DVD uses specifications similar to DVD: the base disc is 0.6 mm thick, and the protective layer is 0.6 mm thick. Blu-ray uses a 1.1 mm base disc, with a protective layer only 0.1 mm thick. This “small” difference means a lot: in Blu-ray, the recording layer is closer to the disc surface. This means the laser has to pass through less material to read from and write to. This in turn makes for a higher NA for the laser lens (see figure *The Effect Of NA On Lens Focusing*). A higher NA, as we said, means a lower track pitch and a smaller pit length. And the lower track pitch and smaller pit length mean more data can be packed onto a BD than an HD-DVD.

So why does HD-DVD use a thicker protective material, and how come a BD won't get scratched easily?

Track Pitch And Pit Length



At left is the pits and bumps on a DVD substrate. The bumps are the areas where there are no pits. The pit length, and the track pitch—the distance between two consecutive rounds of the data spiral—are marked. To right is a high-density (blue laser) DVD; the pit length is smaller, and so is the track pitch